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Assessment on potential security threats and introducing novel data security model in cloud environment

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ABSTRACT

Cloud computing technology is the most important development in engineering. The technology is service-oriented, prioritizing cost reduction, equipment reduction, and payment only for used services. This article reviews current research on security risks and possible solutions for cloud computing. Cloud computing refers to Information systems, network, and device storage are examples of customer-provided services. Because the cloud is operated remotely, it can be accessed from anywhere. Computer service is used by the cloud provider. Unauthorized individuals may access critical information on the cloud, making it insecure. As a consequence, to ensure comprehensive data security, specialist data protection encryption technology is deployed. The topic of data security in cloud computing is discussed in this article. This is a study of cloud data and the security challenges that come with it. To achieve a better result than previous research endeavors with unique constraints, this study used two data encryption and decryption algorithms. The suggested approach is both quicker and easier to use than the RSA-Blowfish method.
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1. Introduction

One of the fastest-growing technologies is cloud computing (CC), with Microsoft Office 365, Gmail, Dropbox, Gmail, and other services being utilized by everyone in a range of sectors [11]. Data kept on servers accessible through the Internet with simple authentication may be accessed from anywhere in the globe [22]. A Level of Service Agreement (SLA) and a dynamic delivery system with several virtual machines are used to maintain communication between the client and the cloud to fulfil and satisfy the client's fluctuating resource demands. Though CC has a lot of well-known benefits, it also has several security flaws. This issue is still difficult, and it must be handled appropriately to benefit from CC services with better performance and efficiency [13]. CC service providers (AWS, Microsoft Azure, Google Cloud, Alibaba etc.) must handle a slew of regulatory challenges to deliver lower-risk CC services. The cloud is a centralised data storage system that allows users to save and access data in the cloud at any time, as well as edit, save, and remove data [20]. The CC idea removes the need to maintain an on-premises data center. The cloud's core concept is to transfer data from a corporation or organization to a distant location on a cloud provider's website [21]. The cloud saves money, reduces overhead, and allows important parts to be implemented more quickly. Variables that contribute to cloud service advantages and variables that demand them and prioritize various firms reveal that 91% of companies in "Europe" and the "United States" have decreased expenses, which is the primary goal for migrating to the CC [15]. In October 2007, Google and IBM partnered together to make CC more mainstream and appealing [18]. "Public cloud", "private cloud", "community cloud", and "hybrid cloud" are some of the several types of CC [16] (see Table 1)

CC is thought to be the simplest network access for mobilizing computer resources in response to the next generation computing model's future need. Virtualization, service-based computing, utility computing, load balancing, multiple leasing, the ability to make payments utilizing computer resources, and the reduction of high capital expenditures and OPEX are just a few of the advancements made possible by CC [1]. While there are several advantages to securing cloud computing data, anonymity, honesty, and confidentiality are three major impediments to mainstream use [2]. Cloud customers need security against tampering and unauthorized

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Table 1
Table 1: Every Round R has four actions (A1, A2, A3, and A4).

A1	"XOR left half of the rth P-array input data (L)"
A2	"Using XORed details for the F-function of Blowfish"
A3	"XOR output of F function by the right half of the data (R)"
A4	"Swap L and R"

access to their confidential information. Internal and external security threats, failures, and cloud service security problems may all occur throughout the cloud infrastructure phase. The group cites wired writer Matt Honon, whose "Gmail", "Twitter", and "Apple" accounts have been falling since the summer of 2012, and who has deleted all of his 18-month-old childhood images [3]. To ensure "data privacy", "privacy protection", and "trust" in the CC environment, data security challenges must be addressed [4].

To ensure privacy, dependability, and trust, many algorithms and protocols (MD5, RSA, PDP, PoR) have lately been developed [5,6]. This research aims to safeguard data against a variety of data security threats, including data privacy, data integrity, and dependency on cloud-based data. The document contains sensitive data encryption, which discourages potential users and businesses from keeping sensitive information in cloud storage services. The essay assists and motivates scientists to investigate security measures that contribute to the cloud environment's dependability [7]. Data security is improved through data authentication at multiple stages. Forensic virtual machines and a variety of encryption methods help to maintain excellent data security, privacy, integrity, and trust.

In this paper Section 2 introduces a brief about CC layers and then Section 3 gives the potential cloud attacks. Section 4 explains the data protection strategy and the proposed methodology explained in Section 5. Section 6 gives the results and discussion. Finally, the work is concluded in Section 7.

2. Cloud computing layer

Most machines in CC use the Internet to access services. Customers benefit from a variety of dependable services provided by cloud computing. CC provides customers with a variety of services like "hardware", "software", "data access", and "capability". CC layers are categorized into three layers: first, CC expansion models, which include "public cloud", "private cloud", "community cloud", and "hybrid cloud"; second, CC expansion models, which include "public cloud", "private cloud", "community cloud", and "hybrid cloud"; and third, CC expansion models, which include public cloud [17]. Self-service on-demand is the second tier of CC functionality, which is dependent on the nature of the CC version and user demand. Then there's the enormous network. Recreate infrastructure, "storage", "platform", and "data resources". Flexible provisioning is made possible by rapid flexibility. SaaS, PaaS, and IaaS are the three CC delivery types. The many levels of CC are shown in Fig. 1.

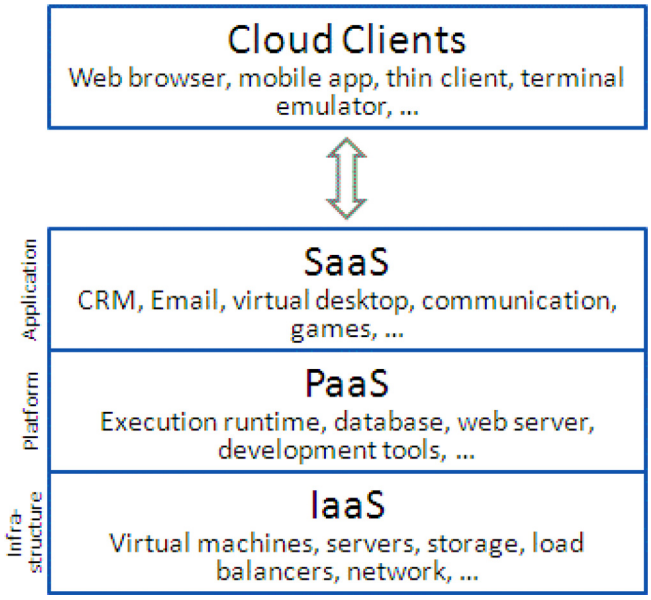


Fig. 1. Various layers of CC.

2.1. CC models

This section discusses the cloud expansion categories in detail, each of which has four CC categories, each of which is described below. Fig. 2 below shows the CC development model in the form of custom graphs [8,9]. Fig. 3 shows the advantages/disadvantages with an example.

2.1.1. Private cloud

This form is only used on behalf of the firm; thus, it is not available to the general public. A private infrastructure offers the cloud for a business or organization's private and specialized purpose, and it is only utilized by the company since no other organization has access to it. Certain CC services, like "dependability", "flexibility", "data security", and "scalability", are among its qualities. For all members, the Enterprise Network is in charge of the data center service provider, which includes virtualization and distributed computing [10,11]. They are fantastic for secrecy and authentication. Eucalyptus, VMware, and IBM Cloud Burst are private cloud vendors.

2.1.2. Public cloud

One of the various growth models, computer infrastructure, and resources publicly accessible on the Internet is the public cloud [17]. A corporation, educator, government agency, or any combination can control, own, and run it. The benefit of being on a big scale is that if the general public has access to it, the provider will provide public cloud services, which may be paid for or given to the client for free [17]. The person who supervises and administers the resources on the service provider's physical location is known as the outsourcer [17]. "Amazon EC2", "Salesforce.com", "Microsoft Azure", and "Google App Engine" are just a few of the cloud vendor services available [11].

2.1.3. The cloud of the community

A community cloud is a hybrid of public and private cloud services that a group of businesses or partners may utilize for both private and public purposes. A community cloud can be on or off campus, and its establishment, maintenance, and operation are the responsibility of one or more organisations, a third party, or a combination of all three [11]. The cloud should only be accessed by a single user group focused on general concerns (e.g., mission, security requirements, policy, etc.).

2.1.4. Cloud hybrid

The cloud framework involves two or three independent cloud networks ("private", "group", or "public") that are uniquely related to structured or proprietary technology (e.g., cloud load-balancing explosion). Private cloud and public cloud are less flexible and powerful than hybrid cloud. The predominance of CC is seen in Fig. 2 [12].

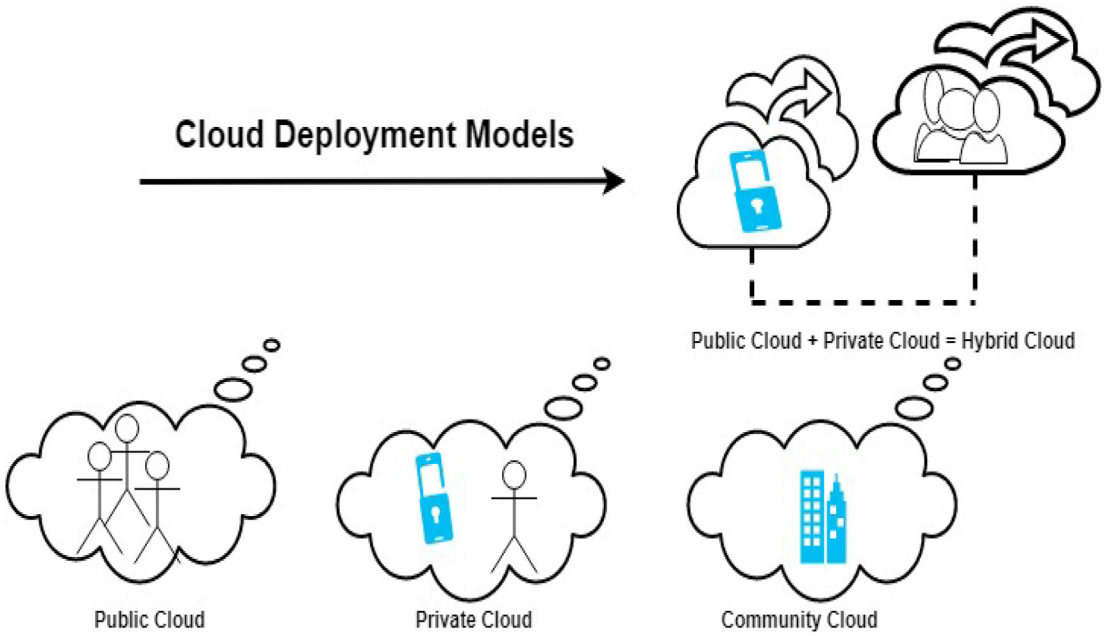


Fig. 2. Cloud computing deployment models.

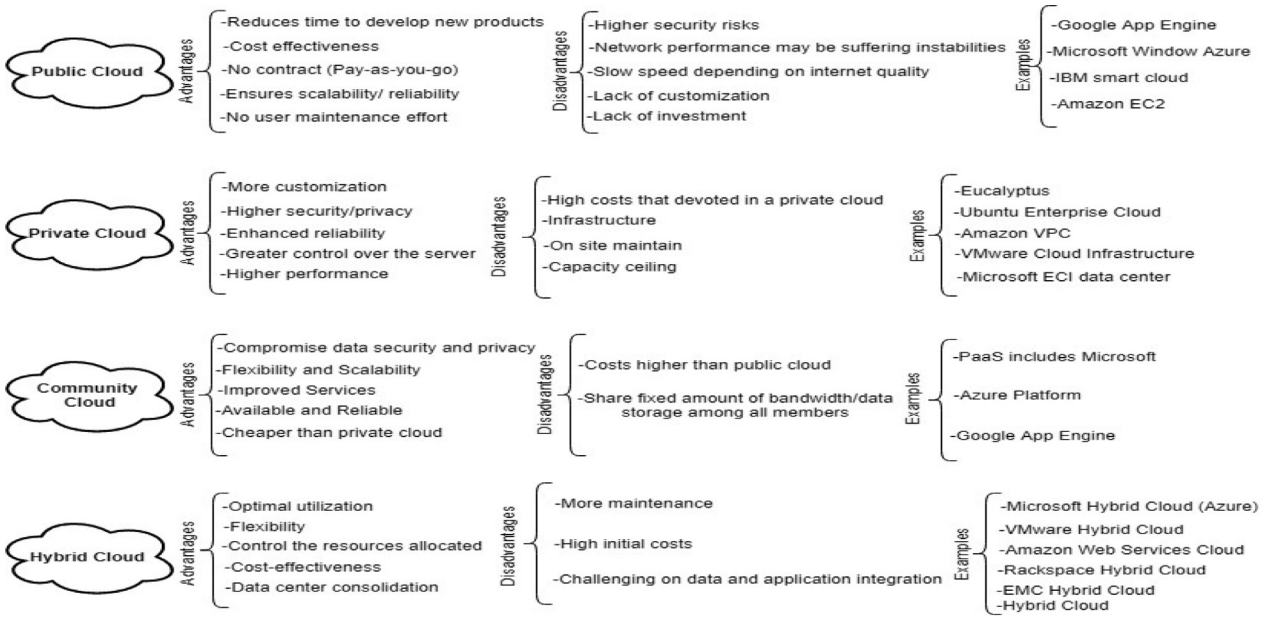


Fig. 3. Cloud deployment models-advantages and disadvantages.

2.1.5. The safety of CC

Security guarantees that the data is not deleted, lost, or infringed upon since only authorized individuals to have access to it. Because data is such an important part of any organization, a loss of "confidentiality", "integrity, and "availability" may have a significant influence on a CC corporation. Data privacy is the core purpose of cloud computing. Encryption and key distribution are used to ensure privacy. The cloud computing key and how it may be obtained and safely sent may have certain concerns. Furthermore, all APIs and service requests from any other application need application keys in Platform as a Service (PaaS). As a consequence, access to these programs and data has to be secured [17]. One of the most important topics in cloud computing security is integrity. Unauthorized access does not affect data stored in the cloud. The system's availability affects its time in the system as well as the lack of disruptions or logs.

2.1.6. Threats to the cloud

This section highlights the most pressing concerns and threats that the cloud computing system faces. Despite advances in research and technology, cloud security still has a lot of flaws. These concerns are shared by a large number of individuals. Security is always a significant concern due to the absence of effective solutions to address vulnerabilities and secure information from any danger such as damage, disclosure, or change. The lack of this attribute, or the existence of a mistake that affects it, may harm computer design, with potentially disastrous moral, personal, and financial consequences [4]. This may have major ramifications,

including data loss. These dangers are one of the most serious security concerns that influence and compromise data [6]. Hackers are attempting to exploit a weakness, according to threats. The Cloud Security Alliance (CSA) [13] lists the names of 12 items.

2.1.7. Breach of personal information

Unauthorized access to and retrieval of data is referred to as data infringement. It is one of the most significant and prestigious species on the planet. You want to eliminate security breaches, such as revealing or selling sensitive and sensitive data, or supplying illicit websites. "Intentional" or "accidental disclosure", "alteration", or "gathering of data", either "directly" or "indirectly", by the user or attacker who is a data breach offender [7]. *Authentication is disabled due to hacked credentials;* Defending against an attacker's incapacity to access solutions and approvals to fill in the holes in authentication management is always tough for enterprises. *The application program interface and hacking interface;* Application program interfaces (APIs) are used by cloud service providers to deliver a range of services to consumers, leading in no complexities and no strong method to blocking or exploiting vulnerabilities, increasing the risk, vulnerability, and exploitability of attackers. The attacker manipulates, resists, listens, and engages in a variety of destructive assaults on the victim [14] exploiting flaws in the system. To get access to systems, attackers take advantage of software flaws or breach the firewall barrier. This is one of the most serious flaws in the cloud's security theft of a bank account. The capacity of an attacker to access data and do operations and activities such as stealing the account and stealing and stealing the email of any account related to a person, organization, or computer, is the method by which an attacker steals an account. Then they grab the login information. The attacker may tamper with and manipulate the data, as well as to conduct phishing, fraud, and software exploitation assaults [15]. *Nefarious insiders;* People have the capacity and authorization to use a firewall or intrusion detection system (IDS) to get access to systems that might affect an existing company and try to hack and acquire access to private and sensitive information.

2.1.8. Advanced persistent threats (APT)

The attacker infiltrates the system and gains access to the infrastructure, allowing him to steal data. Because they utilize complex tactics such as direct hacking, phishing, network infiltration, and vulnerable programming interfaces [16], these sorts of assaults are difficult to detect.

Data loss that is irreversible; Natural calamities such as floods, deliberate or inadvertent data erasure, and human reasons such as viruses or power outages all contribute to data loss. There isn't enough focus. Companies with insufficient previous knowledge and expertise of the services supplied by service providers, as well as those who are unaware of the cloud's effects and hazards.

Violations of cloud services; Customers have unlimited processing and storage options, as well as trial periods during which anyone can start using cloud services, allowing malicious code developers and criminals to use the cloud to carry out unethical and damaging attacks and activities. CC services are provided by "Twitter", "Amazon", and "Facebook". Denial of service (DoS) assaults [17]. By making many requests without replying to repeat services with unanswered requests, an attacker may begin to slow down and finally terminate services by blocking services and blocking UDP floods, SYN floods, ICMP floods, and buffer overflows.

Technology that is shared and common threats; Technology issues occur in a multi-sectoral framework. Shared Infrastructure provides the on-demand service, allowing several customers to utilize the same virtual machine. Security is a big concern in cloud computing, and it manifests itself in a variety of ways, including access management, which is comprised of two main keys. The first is a collection of authentication and identification groups; the second is a hard access strategy. This is the employee's greatest goal value; a substantial proportion of data breaches are caused by human mistakes, and then there are data breaches, which pose a new danger to cloud systems. In other words, a massive volume of data is being exchanged between workers and the cloud system so that attackers may thwart their efforts to do damage. To avoid data breaches, all data going over the network must be encrypted, and SSL/TLS certificates must be installed. Another cloud computing issue is data loss, which occurs when data overflows, making backup difficult and costly. When there are no regular backups and the data is in danger from viruses, malicious assaults, and ransomware, the attacker encrypts the cloud storage, targets the exploits, and demands money to restore the data [17]. Data loss and vulnerability may result from improperly configured cloud storage [18]. Some businesses pre-configure security settings for cloud storage. Others leave public material unencrypted since it is so easy to access it without authorization or request [22].

3. Potential cloud attacks

CC can survive a variety of threats, including hackers and vandals, as well as unrestricted access. There are several potential cloud attacks available in the literature which are given below.

3.1. DoS (Distributed Denial of Service) attacks

CC places a heavy burden on the service through the operating system, making cloud service attacks more vulnerable to rejection. Because the cloud is distributed to a large number of clients, CC places a heavy burden on the service through the operating system, making cloud service attacks more vulnerable to rejection. It begins to supply a significant amount of computational capacity, such as by increasing service states to handle more virtual machines and workload.

3.2. Malware infiltration in the cloud

A malware service application or a cloud-based VM is used to try to compromise a system. Malware operates by listening and adapting. This attack necessitates the adversary creating a virtual machine (a service known as infrastructure) enabled by SaaS or PaaS and uploading it to the cloud as an example of its malware execution unit. Because the cloud system automatically transmits genuine user requests to the malware service execution, it has been hacked.

3.3. Authentication-related attacks

Authentication is one of the most prevalent flaws in hosted services that attackers exploit. Most services now employ username and password authentication, however financial institutions use a range of secondary authentications, such as security questions and biometrics.

3.4. Attacks on the middle man

It is a type of attack in which the attacker listens in on the key exchange between the sender and the recipient. Because the sender and recipient are continuously interacting, the attacker may change his key to anything he wants. The connection is completely in the attacker's control.

4. Data protection strategy

To strengthen and achieve stronger security and security in the cloud, a variety of defense mechanisms should be utilised to control and prevent attacks or data loss. [19]. According to NIST, vendors and companies should use well-known algorithms. The level of protection of the keys and algorithms used will be evaluated on an annual basis. Businesses and organizations use cloud computing. Understanding data-related security measures in a multi-tenant cloud system are crucial. To store keys, it is recommended to use hardware security modules or HSM modules. SaaS, PaaS, and IaaS services are available to end-users in the cloud. In the context of CC, security layers have been introduced into service models [22]. CC includes methods such as privacy utilization, catalog creation of data assets, availability and security, access management device, and authentication to ensure data and privacy requirements. National Institute of Science and Technology express to encrypt and decrypt AES-compliant electronic data. Since data types such as images may be different than content, AES has been found to have a favorable position in RC2, Blowfish, and RC6 over time. In 2005, Sahai and Waters proposed a new type of cipher called a functional cipher, often referred to as a feature-based cipher. Identity-based encryption is Advanced Encryption (IBE). There are two main forms of ciphertext approach and KP-ABE feature-based encryption. ABE is defined as an algorithm in which a user's secret key and any of its encryption methods are affected by a specific attribute, such as "location" or "account type". Assume that the main features of the client fit the cipher. The cipher (CP-ABE) is not decrypted.

5. Methodology of study

5.1. Problem statement

5.1.1. In terms of security and performance, the AES algorithm is inadequate.

The encryption technique gives a tiny variation in time between encoding and decoding since the collector repeats thousands of bits, therefore the difference is visible. Other key emerging uses, such as big data and secure cloud storage, may not be compatible with safe 128-bit AES encryption. As a result, huge data applications may need a complex mathematical and organized algorithm capable of handling a wide variety of business speeds.

5.2. Methodology to be followed

When a corporation or user is kept in the cloud, information can be retrieved from any dispersed or connected cloud service. Here, data storage verification becomes a vital feature of secure communication. The suggested model is the most dependable model, which uses the trusted and safe ciphers AES and RSA to help give additional protection to persons on stored data and therefore increase security via several layers of encryption and encryption procedures.

5.3. RSA algorithm

Ron Rivest, Adi Shamir, and Len Adleman, the algorithm's developers, dubbed it the RSA algorithm in 1977 [RIVE78]. The RSA algorithm is a public-key encryption algorithm that is widely used around the world. It can be used to break up encrypted messages without the need for a secret key.

Both public and digital signatures may use the RSA technique. The factor's complexity has been mostly addressed.

Without exchanging secret keys, Client A delivers a communication to Client B that is encrypted. For the cipher message, only B's public key is used, and B only uses the private key that he knows. The RSA can also be used to sign A with the private key and the message, as well as B with the public key, for authentication.

RSA's algorithm for generating key pairs

Input: length in kbits.
Output: N is the unit of two prime numbers of the same shape (N = pq) not longer than k bits in the RSA prime pair ((N, e), d).
And d is ed um 1.mod $(p-1)(q-1)$. And d is a one-of-a-kind exponent.
Choose an e number from the list 3, 5, 17, 257, 65, 537
Repeat
$p \rightarrow$ GenPrime (k/2)
(pmode) up to 1
Repeat
$q \rightarrow$ GenPrime $(K - K/2)$
(qmode) up to $\neq 1$
$N \rightarrow pq$
$L \leftarrow (p-1)(q-1)$
$d \rightarrow$ modinv (e, L)
Return value (N, e, d)
Encryption: Sender A does the following:-
Get the receiver B (n, e) public key.
The plain text message $1 < m < n$ is defined as a positive integer mm.
Cipher computation c = memodn.
Biki sends ciphertext.
Decoding: Recipient B does the following:-
It uses its private key (n, d) to calculate m = cdmodn.
Extract plain text from the letter mm.

5.4. Blowfish

Blowfish is the asymmetrical black chip that can replace DES for drop encryption. It has a bit size range of 32–448 bits, making it suitable for both domestic and international use. Bruce Schneier designed Blowfish in 1993 as a free and simple alternative to existing encryption techniques. It's been thoroughly examined since then, and it's constantly been regarded as a solid encryption method. Blowfish is a free, patented, and unlicensed encryption algorithm that gives excellent software encryption rates but has yet to be proven practical. Blowfish was created by Schneier as a general-purpose algorithm to replace DES and Aging without the challenges and constraints that other algorithms have. It's a Feistel cipher with 16 rounds and lengthy S chests. This is the same as CAST-128, except with the S squares on the frame fixed. Fig. 4 depicts the blowfish coding procedure. Each line has a 32-bit value. There are five primary ranges: an 18P range (labelled P to prevent misunderstanding) and four 256 S-boxes. The diagram is made up of two sets of subsets.

This F function's 32-bit I/p is split into four 8-bit quadrants and utilized as S-box inputs. S-is available in 8-bit i/p and 32-bit o/p formats. The last 32-bit outputs are XORed and performed at 232. (See the image in the upper right corner.)

Leave the final exchange after the 16th round and XOR L with P18 and R with P17 (resulting in bleach). Except for P1, P2, ... , P18, decoding is identical to cipher and is done in reverse order. Affiliate is difficult since XOR is interchangeable. It's a typical blunder to use reverse cipher order as a decryption technique.

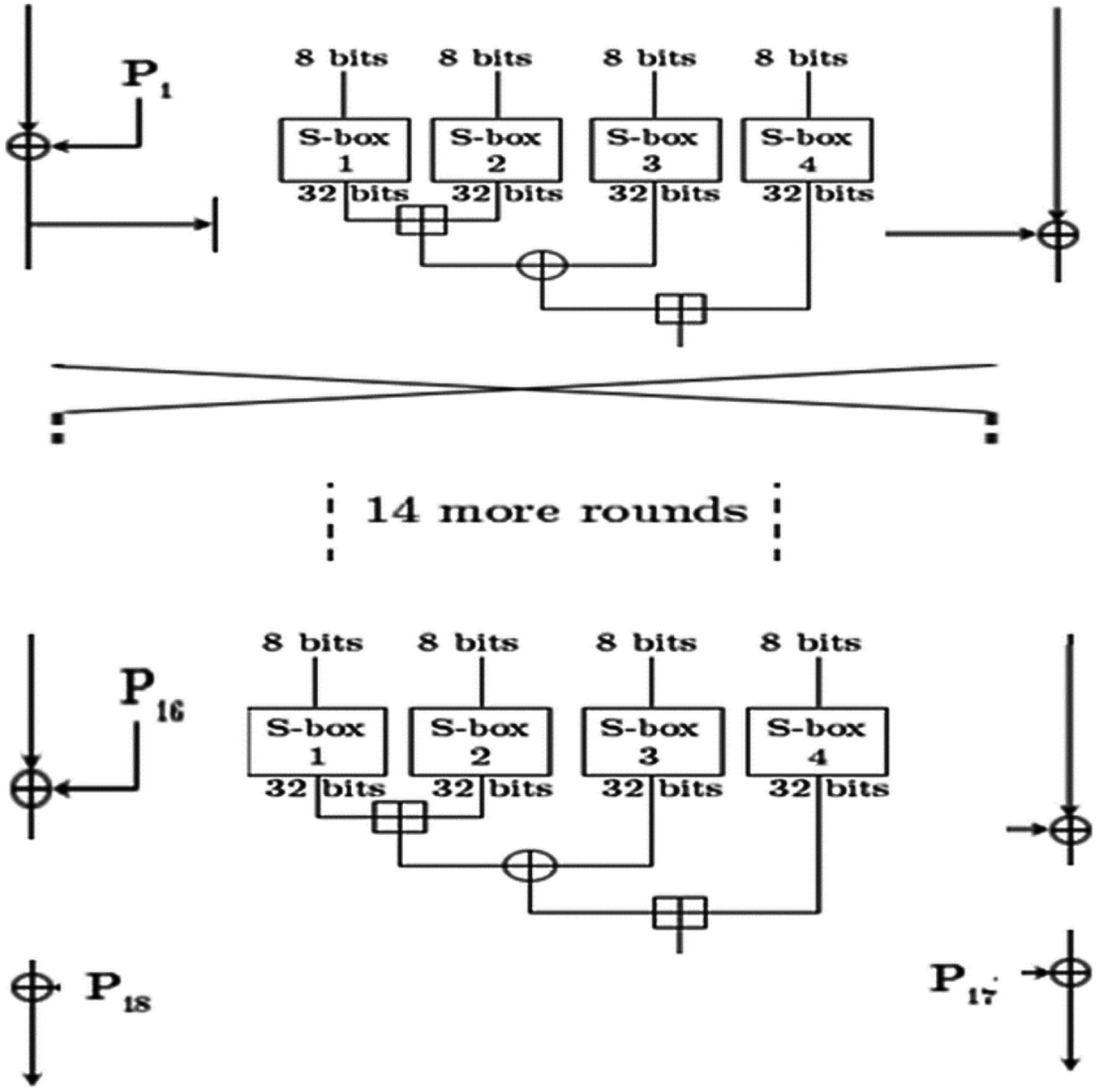


Fig. 4. Feistel structure of Blowfish data flow diagram.

6. Result and discussion

The study was carried out using the Python 3.6 platform. Fig. 5 shows the RSA and Blowfish, two of the newest and most extensively utilized lightweight encryption algorithms for IoT devices.

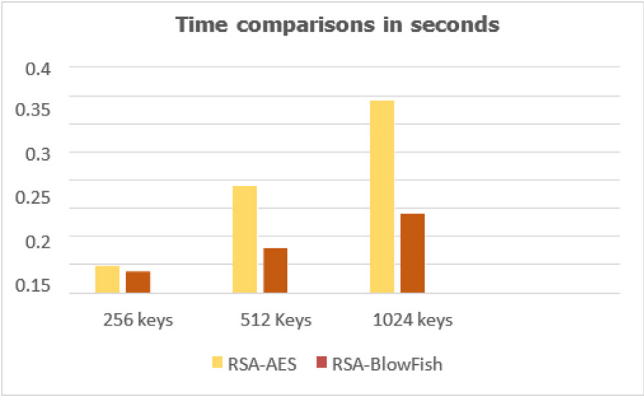


Fig. 5. Comparative study between existing and current research.

Since then, the AES method has been employed in prior works, including the present study. Because the blowfish method outperformed the AES algorithm in previous studies in terms of security, efficiency, and speed, it was utilized in the aforementioned model instead of AES. Blowfish was also shown to be significantly quicker than the AES employed in a prior cloud data security study, but my model AES outperformed the AES in terms of "dependability", "storage space", "data extraction speed", and "time". Blowfish's method is also significantly faster than AES, which is used in my proposed model.

7. Conclusion and future direction

Over time, the latest generation of technologies is evolving to contribute to the growth of the digital world; Among them, CC is one of the main tools that can solve real-world problems. There are several security issues associated with cloud computing. The main objective of this paper is to analyze CC security "threats", "attacks", and "data protection" practices in CC that have identified a key factor in enhancing authentication and access control within the cloud. CC is made more secure by the use of encryption algorithms. Encryption is a technique for securing data by converting it into unreadable or encrypted forms. For the encryption level,

however, there are only a few encryption techniques or algorithms. On one level, any unauthorized individual may simply break the encryption. To offer extra safety for cloud storage, the system employs several layers of encryption and decoding. Because the suggested approach is an encryption and decryption cross-level algorithm. As a result, the data in the proposed work can only be accessed by the authorized user. And if an intruder (unauthorized user) obtains data by accident or on purpose, the data must be decrypted at any level, which is a tough process without a valid key.

CRediT authorship contribution statement

Shahnawaz Ahmad: Data curation, Formal analysis, Conceptualization, Writing – review & editing, Investigation, Methodology. **Shabana Mehfuz:** Data curation, Investigation, Methodology, Validation, Software, Writing – review & editing, Software. **Javed Beg:** Data curation, Visualization, Investigation, Validation, Software.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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